SPC Training Guide - From Operator to SPC Champion

3-level SPC training curriculum | AIAG SPC 2nd Edition

QUALITY HEAD AGENTS | AIAG SPC 2nd Edition | IATF 16949:2016

Level 1 - SPC Awareness (Operators on SPC Lines)

- Duration: 2 hours
- What is a control chart and what the centerline, UCL, LCL mean
- How to plot a reading correctly - subgroup size, timing, recording
- The 4 most important Nelson rules: Rule 1, 2, 3, 4 with examples
- What to do when a point is outside control limits: STOP, CONTAIN, NOTIFY
- What NOT to do: never adjust the process when the chart is in control
- How to fill in the out-of-control log correctly

Level 2 - SPC Practitioner (Engineers, QA Staff)

- Duration: 1 day
- AIAG SPC 2nd Edition framework: variable vs attribute charts
- X-bar & R chart construction: subgroup size selection, control limit calculation using A2, D3, D4, d2
- All 8 Nelson rules - detection and response
- Process capability indices: Cp, Cpk, Pp, Ppk - calculation and interpretation
- Short-run SPC: Z-chart method when insufficient data for traditional limits
- Attribute control charts: p-chart, np-chart, c-chart, u-chart
- SPC plan development: linking Control Plan to SPC chart type and frequency
- PPAP Element 9: initial process study requirements and submission format

Level 3 - SPC Expert / Facilitator (Quality Head / Manager)

- Duration: 2 days + practical assessment
- ANOVA for SPC: partitioning within and between subgroup variation
- Non-normal distributions: transformations, non-parametric capability indices
- Measurement system interaction: MSA-SPC linkage, effect of GRR on observed Cpk
- Customer-specific SPC requirements: Ford Statistical Methods Manual, GM SPC requirements
- Developing plant-wide SPC deployment plan and capability review system
- IATF audit simulation: respond to SPC audit questions
- Facilitating Cpk improvement projects using SPC data